

U-DRIVE Scaffold & Deployment Verification Report

FlyRank AI Internship — Phase 3: Build (Foundations Deliverable)

1. Executive Summary & Core Deliverable URLs

The initial foundation for U-DRIVE has been scaffolded using Next.js App Router, TypeScript, and Tailwind CSS. Continuous Integration and Deployment (CI/CD) have been established via Vercel, ensuring every commit generates a live preview URL. All architecture choices prioritize Server Components by default, reserving Client Components strictly for interactive elements.

Resource Deliverable	Live URL / Repository Reference
GitHub Repository	https://github.com/NusaibaSurovi/U-DRIVE-A-Automobile-E-Commerce-Website-
Vercel Live Preview URL	https://u-drive-automobile.vercel.app (Live Deployment)

2. Application Routing & Component Architecture

Every screen required in the U-DRIVE specification has been mapped to a dedicated App Router path:

Route Path	Screen Name	Component Type	Render Strategy & Purpose
/	Home / Catalog	Server Component	Main landing page showcasing inventory cards, hero banner, and static metadata.
/inventory	Car Inventory	Client Component	Interactive search, category filtering, and price range sliders ('use client').
/cart	Shopping Cart	Client Component	Global cart management leveraging React Context API for dynamic item totals.
/checkout	User Settings/Checkout	Client Component	Interactive checkout form with real-time phone and email field validation.

<code>/api/health</code>	Health-Check Endpoint	Server Route	Renders fetched API status and environment variables verification data.
--------------------------	--------------------------	--------------	---

3. Design Tokens & Design System Integration

The design tokens defined in Week 03 (Identity Kit) have been configured in `tailwind.config.ts` to enforce visual consistency across all routed components:

```
// tailwind.config.ts configuration snippet
colors: {
  primary: '#800040',    // Deep Dark Pink / Plum
  nearBlack: '#111827',  // Body Text / Heading Slate
  nearWhite: '#FAFAFC',  // Background Canvas
  lavender: '#E6E6FA'    // Soft Accent & Focus Ring
}
fonts: {
  heading: ['Montserrat', 'sans-serif'],
  body: ['Inter', 'sans-serif']
}
```

4. Evaluation Criteria Verification

- **Preview URL Build Status:** PASSED — Vercel build succeeds without compilation errors or TypeScript warnings.
- **Routed Placeholders:** PASSED — Every screen from specification exists with root layout and navigation headers.
- **Responsiveness Testing:** PASSED — Layout tested and validated at 375px (Mobile) and 1280px (Desktop) viewports.
- **Repository Security:** PASSED — Zero secrets or API keys stored in Git. Environment variables injected via Vercel dashboard.
- **Health-Check Route:** PASSED — `/api/health` endpoint correctly fetches and renders JSON system status.